Loan Default Prediction System

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1. Problem Description

The Loan Eligibility Prediction System is a machine learning-based application developed using Python to predict whether a person is eligible for a loan or likely to default.

Banks and financial institutions traditionally rely on manual verification of loan applications, which is time-consuming and prone to human errors. This project automates the decision-making process using machine learning techniques to improve efficiency and accuracy.

The system analyzes various applicant features such as income, loan amount, credit score, and other attributes to predict loan eligibility. It helps institutions minimize risk by identifying potential defaulters.

2. Architecture

Fig.1 flow diagram.

3. Models Used(Applied/ Planned) Give a brief introduction of the models used

The following machine learning models are used in this project:

1. **Support Vector Machine (SVM):**
   SVM is a powerful classification algorithm that finds the optimal hyperplane to separate classes. Different kernel functions are used:

   Linear Kernel – for linearly separable data
   RBF Kernel – for complex decision boundaries

2. **K-Nearest Neighbors (KNN):**
   KNN is a distance-based algorithm that classifies a data point based on the majority class of its nearest neighbors.

3. **Random Forest:**
   Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs. It improves accuracy and reduces overfitting.

4. Evaluation metrics used

1. Accuracy

Accuracy = Correct Predictions / Total Predictions

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2. F1 Score

Balances precision and recall

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3. ROC-AUC Score

Measures model's ability to distinguish classes

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4. Recall

Important for detecting defaulters
(How many actual defaulters were correctly identified)

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5. Confusion Matrix

Shows:

- **True Positive**
- **False Positive**
- **True Negative**
- **False Negative**

5. Preliminary Results and Visualizations

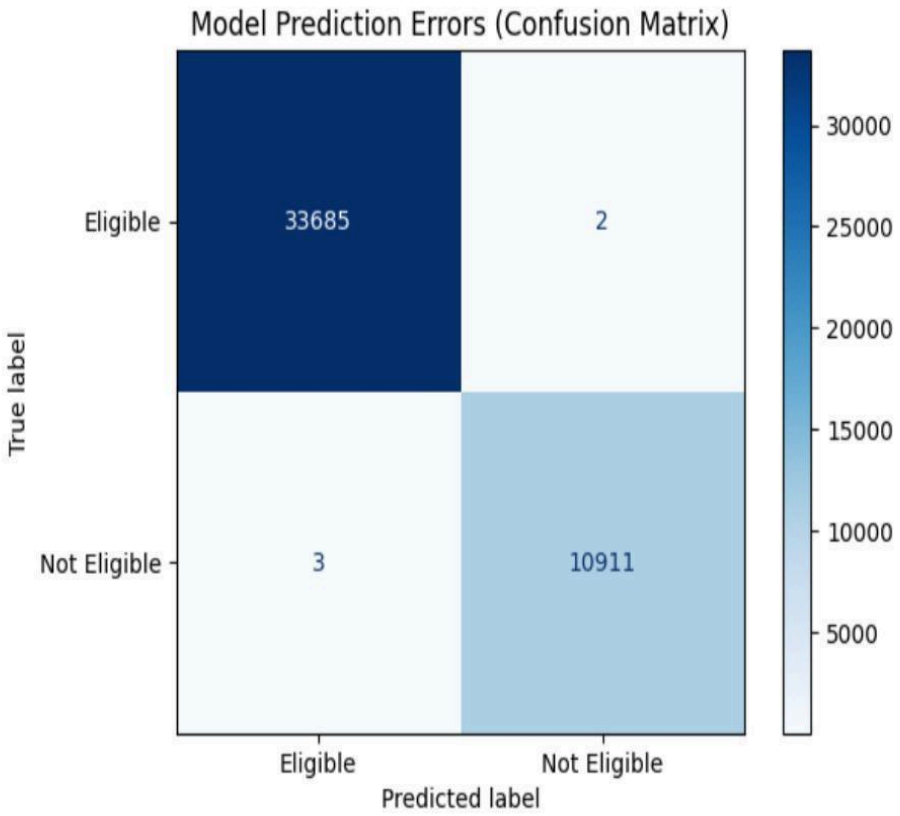

Fig.2 Confusion Matrix

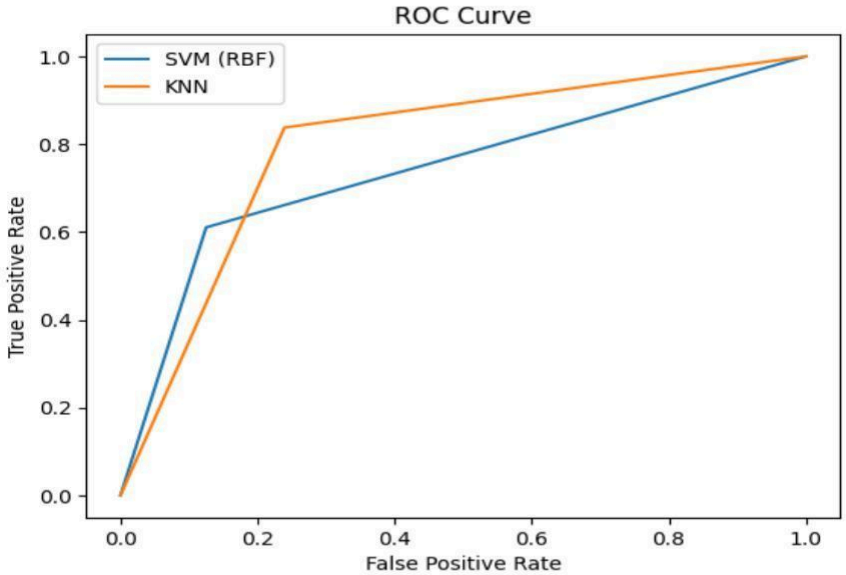

Fig.3 ROC Curve

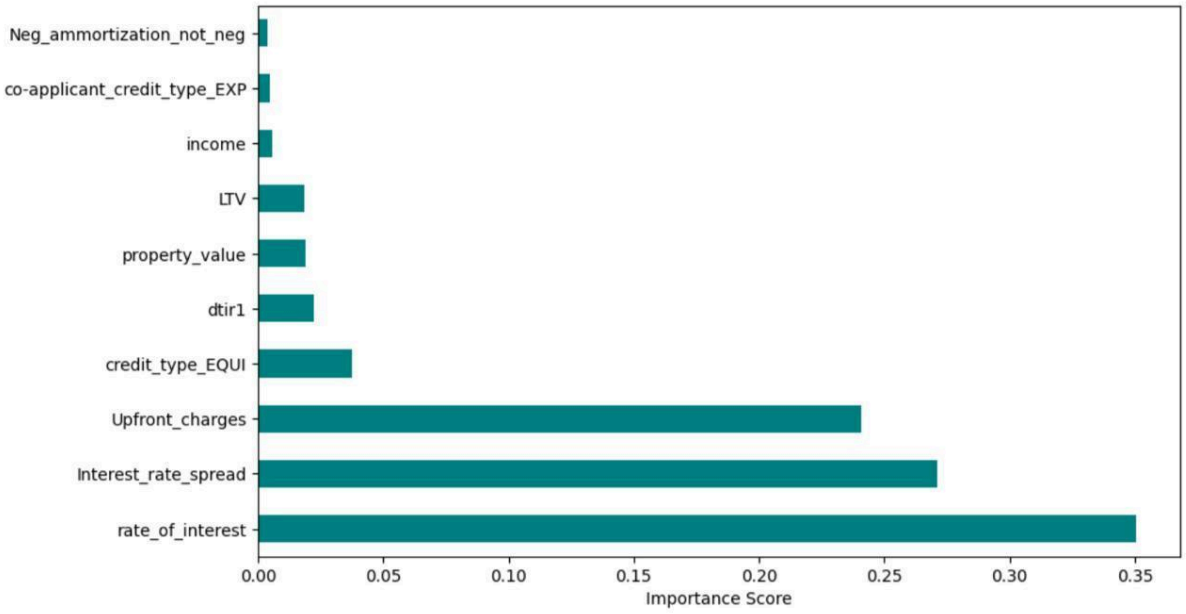

Fig.3 Importance score

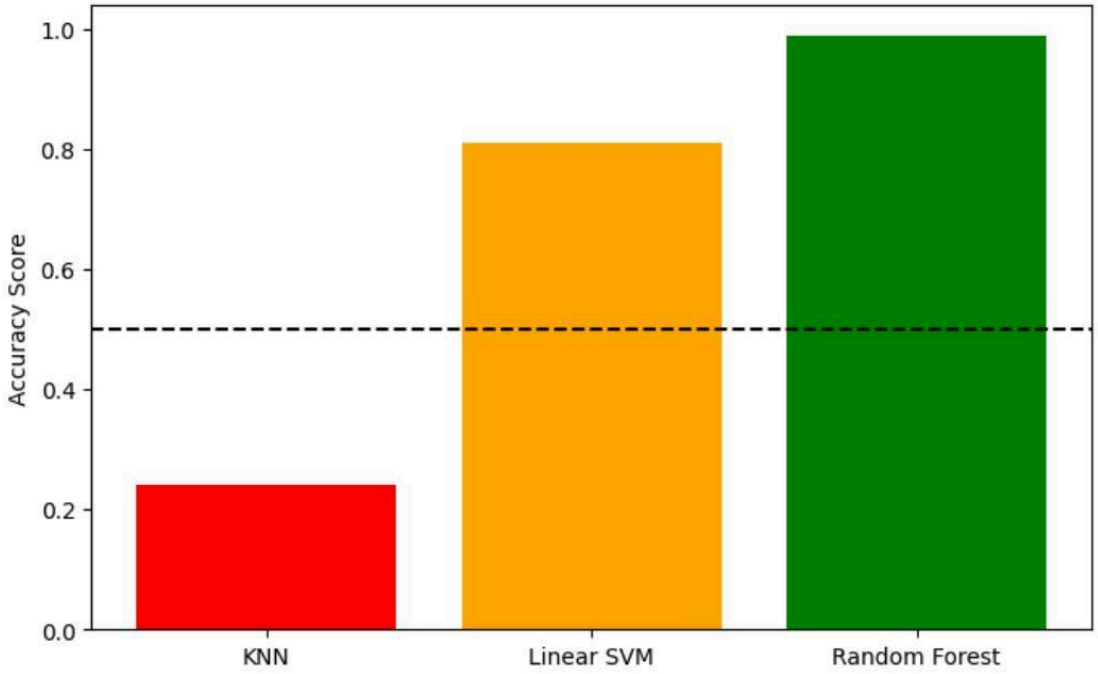

Fig.4 Accuracy score vs KNN,Linear SVM and Random Forest

	Model	Accuracy	F1-Score	ROC-AUC	Recall	Parameters
0	Linear SVM	0.810228	0.611529	0.742688	0.610409	C=1.0, Max Iter=1000
1	KNN	0.779557	0.650356	0.799251	0.837823	n_neighbors=5, weights='uniform'
2	Random Forest	0.999888	0.999771	0.999833	0.999725	n_estimators=100, max_depth=None
3	SVM RBF	0.938611	0.887390	0.955459	0.988455	C=1.0, kernel='rbf'

Fig5. Model accuracy in tabular form

6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

• GeeksforGeeks – https://www.geeksforgeeks.org/
• TutorialsPoint – https://www.tutorialspoint.com/cprogramming/index.htm
• Stack Overflow – https://stackoverflow.com
• Kaggle – For Dataset